

Frequency-Adaptive Sharpness Regularization for Improving 3D Gaussian Splatting Generalization

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Abstract

Despite 3D Gaussian Splatting (3DGS) excelling in most configurations, it lacks generalization across novel viewpoints in a few-shot scenario because it overfits to the sparse observations. We revisit 3DGS optimization from a machine learning perspective, framing novel view synthesis as a generalization problem to unseen viewpoints — an underexplored direction. We propose Frequency-Adaptive Sharpness Regularization (FASR), which reformulates the 3DGS training objective, thereby guiding 3DGS to converge toward a better generalization solution. Although Sharpness-Aware Minimization (SAM) similarly reduces the sharpness of the loss landscape to improve generalization of classification models, directly employing it to 3DGS is sub-optimal due to the discrepancy between the tasks. Specifically, it hinders reconstructing high-frequency details due to excessive regularization, while reducing its strength leads to under-penalizing sharpness. To address this, we reflect the local frequency to set the regularization weight and the neighborhood radius when estimating the local sharpness. It prevents floater artifacts in novel viewpoints and reconstructs fine details that SAM tends to oversmooth. Across datasets with various configurations, our method consistently improves a wide range of baselines. Code will be available at <https://bbangsik13.github.io/FASR>.

1. Introduction

Reconstructing 3D scenes from multi-view 2D images has been a long-standing problem of interest. 3D Gaussian Splatting (3DGS) [23] achieves photo-realistic fidelity in novel-view synthesis with real-time rendering. However, it requires densely captured input views, which are costly to obtain. In sparse-view settings, they often overfit to training views, resulting in poor generalization to novel viewpoints with unresolved details and floating artifacts. To improve

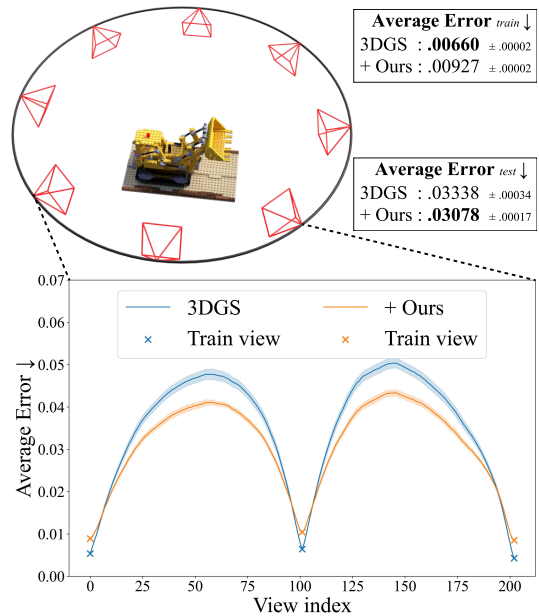


Figure 1. **Overview.** Our proposed optimization algorithm improves generalization. Given 8 training views rendered from the lego scene in Blender synthetic dataset [43], our method maintains low Average Error [46] across interpolated novel-views, whereas the baseline exhibits overfitting. Plots are means and standard deviations over 10 runs.

generalization, previous approaches have adopted various strategies, such as integrating geometric priors from depth and flow estimators [8, 70], and leveraging dense correspondence prediction models [20]. Although these methods show effectiveness, research on improving the 3DGS optimization itself has not been thoroughly explored.

In this paper, we propose an optimization algorithm for 3DGS variants to improve quality in novel viewpoints, i.e., generalization (Fig. 1). Our motivation is to optimize the model toward the flat minima in the loss landscape, ensuring

better generalization. As illustrated in Fig. 2, sharp minima¹ exhibit a large gap between train and test loss, whereas flat minima show a much smaller gap, indicating better generalization. For example, floaters are invisible in training views but critically emerge in novel viewpoints; i.e., the Gaussians overfit to the training views. This large generalization gap implies a sharp minimum, i.e. loss dramatically changes even with slight perturbations on the model parameters².

Although Sharpness-Aware Minimization (SAM) [12] theoretically and experimentally demonstrates that pursuing flatter minima leads to improved generalization of classification networks, *achieving flat minima does not always guarantee better generalization in reconstruction tasks* because the importance of sharpness for accurate reconstruction varies across pixels. In particular, model parameters representing high-frequency details (e.g., edges) inherently induce a sharp loss landscape; even small changes in a well-fitted Gaussian cause a drastic change in the loss. In contrast, for parameters representing low-frequency regions, the loss changes more gradually, favoring flat minima. In this sense, finding a flat minimum for all Gaussians is problematic; sharpness in high-frequency areas is desirable for accurate reconstruction, flatter minima in low-frequency regions are preferable for better generalization.

To this end, we propose Frequency-Adaptive Sharpness Regularization (FASR), an optimization algorithm that penalizes local sharpness, where both the neighborhood radius used for its computation and the weight are set in inverse proportion to the frequency, *encouraging a flatter loss landscape while retaining the sharpness required for fine details*. We show that the enhancement of our method is complementary to prior methods, thereby providing additional performance gains across them.

Our contributions are significant as follows:

- To the best of our knowledge, we present the first fundamental investigation into the relationship between loss landscape and generalization in novel-view synthesis.
- Consequently, we propose an optimization algorithm for 3DGS by reformulating SAM in a frequency-adaptive manner, overcoming the limitation of SAM, which over-smooths high-frequency details.
- Our method is versatile to improve a wide range of baseline methods on various datasets.

¹We omit “local” for brevity.

²Camera perturbations can be interpreted as Gaussian extrinsic parameter perturbations, e.g., parallel movement of all Gaussians is equivalent to moving the camera.

³Figure is reproduced from Keskar et al. [24]. As empirically shown in Izmailov et al. [18], the test loss landscape tends to shift relative to the train loss landscape, while Liu et al. [39] showed that the loss landscape differs with the imbalance level of the dataset.

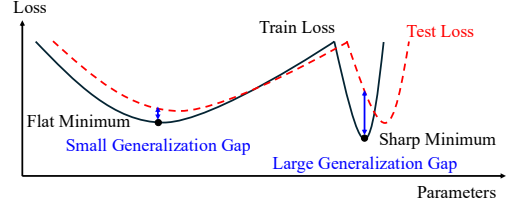


Figure 2. **Conceptual 1D Loss Landscape of Flat and Sharp Minima**³. Flat minimum better generalize than sharp minimum.

2. Related work

2.1. Sparse-view reconstruction

Reconstructing scenes from highly sparse inputs remains a fundamental challenge, as limited supervision often leads models to overfit training views, resulting in insufficient generalization in unseen views. Early efforts sought to extend neural radiance fields (NeRFs) [43] by incorporating additional regularizations or auxiliary cues. For example, these studies [9, 46, 56, 64] introduce geometry- or depth-based constraints and frequency regularization to alleviate the underconstrained nature of sparse-view optimization. Although these approaches demonstrate partial effectiveness, they nonetheless inherit the slow training and rendering processes of NeRF.

Recent studies have shifted to 3D Gaussian Splatting (3DGS) [23], aiming for both real-time rendering efficiency and sparse-view fidelity. These studies [8, 20, 32, 62, 70, 72] leverage external priors such as depth [6, 48], correspondence [31], or flow [51], while another focuses on prior-free regularization of the representation itself [47, 67, 69]. Specifically, CoR-GS [67] optimizes two fields and reduces their disagreements; SE-GS [69] randomly perturbs one field based on rendered image uncertainty. DropGaussian [47] randomly drops Gaussians during training. These methods are essentially stochastic ensemble-like regularizations; their improvement reduces when different subsets converge similarly. In contrast, our method encourages flat minima by adversarially finding the local maxima, leading to more effective and stable generalization.

Our approach is complementary: rather than altering the representation or adding priors, we fundamentally guide 3DGS to converge on a general solution in under-constrained sparse supervision.

2.2. Flatness and generalization

The relationship between the flatness of the loss landscape and model generalization has been extensively investigated in prior research. Keskar et al. [24] shows that converging to sharp minima leads to poor generalization, while Jiang et al. [22] identifies that sharpness is one of the most correlated indicators of generalization. Subsequent work [7, 18] demonstrated that averaging parameters along training tra-

jectories can lead to flatter minima with better generalization.

Beyond averaging strategies, Sharpness-Aware Minimization (SAM) [12] explicitly estimates and reduces sharpness, inspiring subsequent work [2, 35, 44, 53] to analyze and improve upon its formulation. Moreover, some works have analyzed the accuracy of estimated sharpness as a measure of generalization. They show that sharpness changes with parameter rescalings [10], and the appropriate sharpness estimation differs across different training setups and tasks [3], which hinders the correlation between sharpness and generalization. Consequently, recent work introduces normalization and invariant formulations to appropriately estimate sharpness, enabling a more reliable link between sharpness and generalization [16, 19, 27, 29, 55]. On the other hand, some works report counterexamples where sharper models generalize well, indicating that flatter minima are not always the optimal strategy for generalization [4, 10, 59].

Building on this perspective, we revisit the loss landscape sharpness in the context of the reconstruction task. We hypothesize that the optimal sharpness is not uniform but varies spatially, depending on the local frequency content of the signal. Therefore, instead of pursuing a universally flat minimum, we introduce a frequency-adaptive optimization strategy that appropriately estimates sharpness for the reconstruction task and allows sharp minima where they benefit high-frequency details.

2.3. Reconstructing with perturbation

Random perturbation has been introduced into radiance field training for two primary purposes.

Unlike our approach, which aims to improve generalization, some studies employ perturbation for uncertainty quantification rather than relying on a deterministic approach. Stochastic or Bayesian formulations of NeRF have explicitly model radiance or density distributions [30, 50]. Subsequent work perturbs trained models to estimate epistemic uncertainty [15]. Similar ideas have been extended to 3DGS, where Gaussian parameters are sampled from learned distributions to render both images and calibrated uncertainty maps [1].

Another works exploit stochasticity to better optimization. These approaches inject noise into Gaussian parameters or query locations to escape poor local minima [13, 26] or enable a coarse-to-fine training effect [37]. Related approaches also perturb the sampling policy itself for acceleration, encouraging the model to focus on under-trained regions and thereby improving convergence [25].

While these stochastic approaches can improve convergence, they may also introduce unexpected artifacts (Appendix C provides more details). Therefore, we provide a more fundamental alternative by explicitly encouraging flat

minima in an adversarial manner.

3. Method

We first provide preliminary on SAM [12] (Sec. 3.1). Then, we explore applying SAM to 3DGS (Sec. 3.2). Finally, Sec. 3.3 presents our frequency-adaptive sharpness regularization.

3.1. Preliminary: Sharpness-Aware Minimization

Sharpness-Aware Minimization (SAM) [12] is an optimization method that improves generalization by encouraging solutions to lie in flat regions of the loss landscape. Along with minimizing the empirical loss at a parameter point $\mathbf{w}$, SAM estimates the loss sharpness within a neighborhood of radius ρ . The optimization then jointly minimizes both the empirical loss and this estimated sharpness:

$$L^{\text{SAM}} = \overbrace{\max_{\|\epsilon\|_2 \leq \rho} L(\mathbf{w} + \epsilon) - L(\mathbf{w})}^{\text{loss sharpness}} + \overbrace{L(\mathbf{w})}^{\text{empirical loss}} \quad (1)$$

$$= \underbrace{\max_{\|\epsilon\|_2 \leq \rho} L(\mathbf{w} + \epsilon)}_{\text{worst-case loss}},$$

which is equivalent to the worst-case loss. In practice, it is approximated via first-order Taylor expansion for the loss function around the current parameters, *perturbing the parameters* in the gradient direction with magnitude ρ :

$$\hat{\epsilon}(\mathbf{w}) \triangleq \arg \max_{\|\epsilon\|_2 \leq \rho} L(\mathbf{w} + \epsilon) \approx \rho \cdot \frac{\nabla_{\mathbf{w}} L(\mathbf{w})}{\|\nabla_{\mathbf{w}} L(\mathbf{w})\|_2}. \quad (2)$$

Then, the model parameters $\mathbf{w}$ are updated via Stochastic Gradient Descent (SGD) [45] step, with the gradient computed at the estimated local maximum $\mathbf{w} + \hat{\epsilon}(\mathbf{w})$:

$$\mathbf{w} \leftarrow \mathbf{w} - \lambda_{\text{lr}} \cdot \nabla_{\mathbf{w}} L(\mathbf{w})|_{\mathbf{w} + \hat{\epsilon}(\mathbf{w})},$$

where λ_{lr} is a learning rate. Supporting the intuition of Fig. 2, this method improves generalization by tightening the generalization bound based on sharpness derived from the PAC-Bayesian framework [11, 41, 49]:

Theorem 1. *For any $\rho > 0$, with training set $\mathcal{S}$ from data distribution $\mathcal{D}$,*

$$L_{\mathcal{D}}(\mathbf{w}) \leq \max_{\|\epsilon\|_2 \leq \rho} L_{\mathcal{S}}(\mathbf{w} + \epsilon) + h(\|\mathbf{w}\|_2^2 / \rho^2),$$

where $h : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is a strictly increasing function (under some technical conditions on $L_{\mathcal{D}}(\mathbf{w})$).

In practice, ρ is a hyperparameter.

The sharpness can be interpreted as a regularization term, as in Weighted SAM (WSAM) [65]:

$$L^{\text{WSAM}} = L(\mathbf{w}) + \frac{\gamma}{1 - \gamma} \left[\max_{\|\epsilon\|_2 \leq \rho} L(\mathbf{w} + \epsilon) - L(\mathbf{w}) \right] \quad (3)$$

$$= \frac{1 - 2\gamma}{1 - \gamma} L(\mathbf{w}) + \frac{\gamma}{1 - \gamma} \max_{\|\epsilon\|_2 \leq \rho} L(\mathbf{w} + \epsilon).$$

Algorithm 1 Applying SAM to 3DGS

Input: Multi-view images $\tilde{I}^v$, where camera $v \in \mathcal{V}$
Output: Optimized $\mathcal{G} = (\mu_i, \mathbf{q}_i, \mathbf{s}_i, \sigma_i, \mathbf{Y}_i^{\text{DC}}, \mathbf{Y}_i^{\text{AC}})$

```

1: while  $\mathcal{G}$  not converged do
2:    $L \leftarrow \text{Loss}(\text{Render}(\mathcal{G}, v), \tilde{I}^v)$  // Get loss
3:   for all  $\theta \in \mathcal{G}$  do
4:      $\hat{\theta} \leftarrow \theta + \rho_{\theta} \frac{\nabla_{\theta} L}{\|\nabla_{\theta} L\|_2}$  // Ascent step
5:   end for
6:    $\hat{\mathcal{G}} \leftarrow (\hat{\theta} \mid \theta \in \mathcal{G})$  // Local maximum
7:    $\hat{L} \leftarrow \text{Loss}(\text{Render}(\hat{\mathcal{G}}, v), \tilde{I}^v)$  // Get loss
8:    $\mathcal{G} \leftarrow \mathcal{G} - \text{Adam}(\nabla_{\hat{\mathcal{G}}} \hat{L})$  // Descent step
9: end while
  
```

When the weight hyperparameter γ is set to 0, the sharpness term vanishes and the optimization minimizes the empirical loss only. When $\gamma = 0.5$, WSAM is equivalent to SAM (Eq. (1)), while for $0.5 < \gamma < 1$, the sharpness term is emphasized. Accordingly, in the extended Theorem 1, $\max_{\|\epsilon\|_2 \leq \rho} L_{\mathcal{S}}(\mathbf{w} + \epsilon) = L_{\mathcal{S}}^{\text{SAM}}$ becomes $L_{\mathcal{S}}^{\text{WSAM}}$, and the function h is modified.

See Foret et al. [12] and Yue et al. [65] for the full theorem statement and proof.

3.2. Applying SAM to 3DGS

Let $\mathcal{G}_i$ be the i -th Gaussian, defined as $\mathcal{G}_i = (\mu_i, \mathbf{q}_i, \mathbf{s}_i, \sigma_i, \mathbf{Y}_i^{\text{DC}}, \mathbf{Y}_i^{\text{AC}})$, where $\mu, \mathbf{q}, \mathbf{s}, \sigma$, and $\mathbf{Y}$ represent the mean, rotation, scale, opacity and spherical harmonics (SH) coefficients, respectively, with $\mathbf{Y}_i^{\text{DC}}$ and $\mathbf{Y}_i^{\text{AC}}$ corresponding to the DC and AC terms of SH.

As in the 3DGS [23] pipeline, we render 3D Gaussians $\mathcal{G} = (\mathcal{G}_i)_{i=1}^N$ from the camera view v and compute the loss L (Algorithm 1.2). The gradient of this loss guides the parameter to its local maximum (Algorithm 1.3-6). Similar to the SAM pipeline, we then evaluate the loss $\hat{L}$ at the local maximum and use this gradient to update the original parameters via the Adam optimizer [28] (Algorithm 1.7-8).

However, this direct application often leads to suboptimal performance as shown in Sec. 4.3.1. Unlike the classification task, in the reconstruction task, the curvature of the landscape is highly correlated with the image frequency. Loss in high-frequency regions requires sharp minima for accurate reconstruction, making it sensitive to perturbations, whereas low-frequency regions require flat minima and are less sensitive to them. Due to this differing sensitivity, a fixed ρ_{θ} is problematic. In high-frequency regions, it leads to inaccurate sharpness estimation by causing large first-order approximation errors (Eq. 2). Conversely, in low-frequency regions, the perturbation is too weak to be meaningful, limiting the improvement of SAM. Moreover, SAM penalizes estimated sharpness with the fixed regularization weight $\gamma = 0.5$. It over-penalizes the high-frequency re-

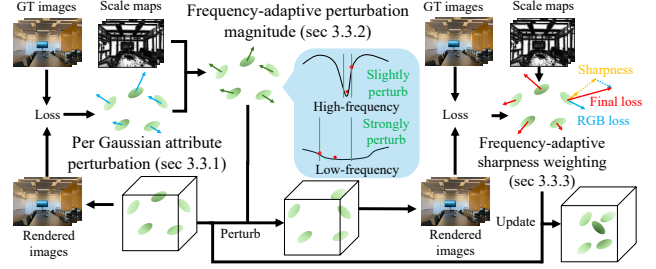


Figure 3. Overview of our proposed method.

gions, which must remain sharp to preserve details. Concurrently, it under-penalizes the low-frequency regions, failing to sufficiently improve generalization.

Key hypothesis. Based on this intuition, we hypothesize that *for each Gaussian, the optimal perturbation magnitude ρ_{θ} and regularization weight γ vary in correlation with image frequency*, which effectively tightens the WSAM-extended version of the generalization bound in Theorem 1.

3.3. Frequency-Adaptive Sharpness Regularization

To address the limitation of SAM, we introduce Frequency-Adaptive Sharpness Regularization (FASR). As illustrated in Fig. 3, we first perturb the per-Gaussian attribute (Sec. 3.3.1). Then we adjust the perturbation magnitude and the regularization weight of each Gaussian⁴ (Secs. 3.3.2 and 3.3.3) using the precomputed scale map Γ^v at view v , which is inversely proportional to the local frequency (detailed in Appendix E).

3.3.1. Per-Gaussian attribute perturbation

Unlike neural networks, 3DGS is an explicit model, which allows us to associate the local frequency of each pixel with its corresponding Gaussian. Before applying frequency adaptivity, we begin by perturbing the per-Gaussian attribute $\theta_i \in \mathcal{G}_i$ rather than the entire parameter set. Accordingly, Algorithm 1.4 becomes

$$\hat{\theta}_i \leftarrow \theta_i + \rho_{\theta} \frac{\nabla_{\theta_i} L}{\|\nabla_{\theta_i} L\|_2}. \quad (4)$$

This step has the advantage of estimating more accurate sharpness by normalizing the gradients of the per-Gaussian attributes. Specifically, it mitigates the first-order approximation error of Gaussians with large gradients.

3.3.2. Frequency-adaptive perturbation magnitude

Going a step further, we adapt the perturbation magnitude according to the local image frequency. Specifically, for each Gaussian, we query the value of the optimal scale map

⁴We apply this adaptivity to the mean μ_i , rotation $\mathbf{q}_i$, and scale $\mathbf{s}_i$, which are geometric attributes of the Gaussian.

Table 1. **Quantitative comparison on LLFF and MipNeRF-360 dataset.** Our method improves baselines across the board.

Method	LLFF (3 views)			MipNeRF-360 (12 views)		
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
3DGS [ACM ToG’23]	19.810 $\pm$.339	.6790 $\pm$.0078	.2145 $\pm$.0065	18.903 $\pm$.179	.5499 $\pm$.0036	.3734 $\pm$.0042
+ Ours	20.783 $\pm$.300	.7197 $\pm$.0032	.1965 $\pm$.0034	19.303 $\pm$.185	.5622 $\pm$.0051	.3552 $\pm$.0047
CoR-GS [ECCV’24]	20.185 $\pm$.142	.7015 $\pm$.0040	.2029 $\pm$.0035	19.515 $\pm$.243	.5733 $\pm$.0049	.3741 $\pm$.0066
+ Ours	20.862 $\pm$.154	.7283 $\pm$.0042	.1932 $\pm$.0030	19.805 $\pm$.233	.5833 $\pm$.0058	.3681 $\pm$.0069
DropGaussian [CVPR’25]	20.461 $\pm$.212	.7070 $\pm$.0045	.2064 $\pm$.0047	19.514 $\pm$.199	.5722 $\pm$.0042	.3657 $\pm$.0036
+ Ours	20.853 $\pm$.227	.7295 $\pm$.0045	.1969 $\pm$.0040	19.625 $\pm$.254	.5750 $\pm$.0053	.3627 $\pm$.0051
NexusGS [CVPR’25]	21.048 $\pm$.049	.7382 $\pm$.0008	.1776 $\pm$.0009	18.506 $\pm$.098	.5222 $\pm$.0031	.3587 $\pm$.0021
+ Ours	21.348 $\pm$.078	.7511 $\pm$.0012	.1714 $\pm$.0011	18.736 $\pm$.103	.5316 $\pm$.0034	.3522 $\pm$.0024
SE-GS [ICCV’25]	20.725 $\pm$.217	.7203 $\pm$.0049	.1861 $\pm$.0058	19.931 $\pm$.288	.5930 $\pm$.0063	.3702 $\pm$.0054
+ Ours	21.141 $\pm$.223	.7403 $\pm$.0042	.1803 $\pm$.0038	20.135 $\pm$.232	.5960 $\pm$.0059	.3644 $\pm$.0052

at the projected center coordinate (x, y) on the 2D camera plane, $\gamma_i \leftarrow \Gamma^v(x, y)$. Since this value is obtained in the 2D image space, we adjust it for the Gaussian mean μ_i and rotation $\mathbf{q}_i$ by multiplying the rendered depth at the same location, $d_i \leftarrow D^v(x, y)$, and dividing by the focal length f , so that Gaussians located farther from the camera v are perturbed more strongly in 3D space. Then, the resulting value is multiplied by a predefined neighborhood radius to determine the final perturbation. Therefore, instead of Eq. (4) Algorithm 1.4 becomes

$$\hat{\theta}_i \leftarrow \theta_i + \gamma_i \cdot \frac{d_i}{f} \cdot \rho_{\theta} \frac{\nabla_{\theta_i} L}{\|\nabla_{\theta_i} L\|_2}. \quad (5)$$

As a result, the perturbation magnitude is adaptive to the frequency of the corresponding Gaussian. This adaptation mitigates the first-order approximation error of Gaussians in high-frequency regions and perturbs strongly in low-frequency regions. Consequently, estimation of sharpness is more appropriate, thus improving generalization.

3.3.3. Frequency-adaptive sharpness weighting

Finally, inspired by WSAM [65], we adapt the regularization weight according to the frequency of each Gaussian, where the frequency is defined the same as in Sec. 3.3.2. From Eq. (3), the final gradient is a weighted combination of the gradient at the original point and the gradient at the perturbed point:

$$\nabla_{\hat{\theta}_i} \hat{L} \leftarrow \frac{1 - 2\tilde{\gamma}_i}{1 - \tilde{\gamma}_i} \nabla_{\theta_i} L + \frac{\tilde{\gamma}_i}{1 - \tilde{\gamma}_i} \nabla_{\hat{\theta}_i} \hat{L}. \quad (6)$$

Here, $\tilde{\gamma}_i \leftarrow 0.95 \cdot \gamma_i / \gamma_{\max}$, where $\gamma_{\max}$ is the maximum candidate scale of the LoG kernel, and 0.95 is determined heuristically. We then update the parameter at the original point using this weighted gradient (Algorithm 1.8).

As a result, the sharpness penalty is reduced in high-frequency regions, preserving fine details, while applying a

stronger penalty in low-frequency regions to facilitate better regularization.

4. Experiments

4.1. Experimental settings

Dataset. We evaluate our method on the LLFF [42] and MipNeRF-360 [5] datasets, following the experimental settings as previous works [47, 67, 70, 72], where the input resolution is $8\times$ downsampled, and 3 and 12 input views are split for LLFF and MipNeRF-360, respectively.

Implementation. We selected the publicly available 3DGS [23] and its follow-up work as baselines. Specifically, we selected the state-of-the-art NexusGS [70], which leverages foundation models, and CoR-GS [67], DropGaussian [47], and SE-GS [69], which do not.

Metrics. We use PSNR, SSIM [58], and LPIPS [68] as evaluation metrics, where LPIPS is computed with a VGG network [52]. Additionally, we use Average Error (AVGE) [46], which is the geometric mean of PSNR, SSIM, and LPIPS. Considering the randomness of 3DGS, we conduct experiments with multiple runs.

4.2. Reconstruction quality

As shown in Tab. 1, our method achieves clear gains in PSNR, SSIM, and LPIPS across the LLFF and MipNeRF-360 datasets. When integrated into 3DGS and its variants, these improvements are achieved without any architectural changes or additional priors, enhancing generalization to unseen viewpoints. Fig. 4 show visual comparisons. Applying our method consistently improves the baselines by correcting geometric inaccuracies and reducing floating artifacts in novel-views. These results reveal that our proposed optimization algorithm can be seamlessly integrated

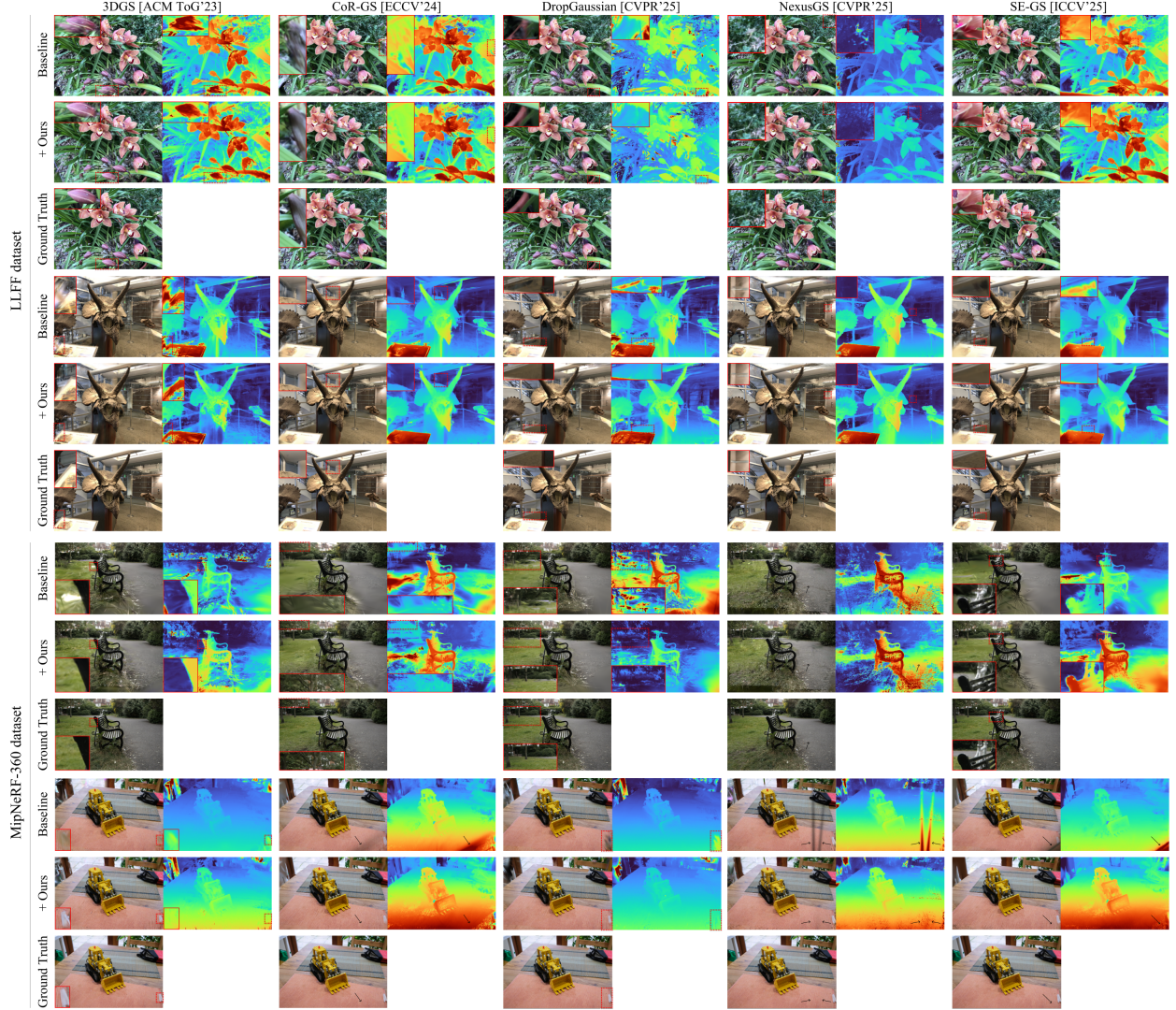


Figure 4. **Qualitative comparison on LLFF and MipNeRF-360 dataset.** Our method consistently improves reconstruction quality.

into current and future 3DGS-based frameworks, providing complementary enhancements.

4.3. Analysis

4.3.1. Ablation study

Directly applying SAM to 3DGS leads to degraded performance. This naive approach strongly perturbs Gaussians with large gradients, resulting in blurry reconstructions (Fig. 5, 6th column). Moreover, it increases the first-order approximation error, hindering the optimization process of SAM (Tab. 2, 7th row). Per-Gaussian perturbation, normalizes perturbation magnitude. Thus, it mitigates this issue, leading to improvements in some metrics (Tab. 2, 6th row), but the results remain blurry (Fig. 5, 5th column).

Frequency-adaptive sharpness weighting preserves sharpness in high-frequency details. Meanwhile, frequency-

adaptive perturbation magnitude enables more faithful estimation by adaptively adjusting perturbation magnitude. Applying either component individually shows performance gains (Tab. 2, 4th and 5th rows). However, removing either component degrades performance (Fig. 5, 4th and 3rd columns). Using both achieves a better generalization, balancing between sharpness reduction and detail preservation (Fig. 5, 2nd column and Tab. 2 3rd row).

4.3.2. Loss landscape visualization

To analyze the convergence behavior of our method, we visualize the reconstruction loss landscape and the corresponding optimization trajectories. We first train 3DGS for 5k iterations. Subsequently, we continue training for an additional 5k iterations with the densification process disabled, under three distinct settings: 3DGS optimization,

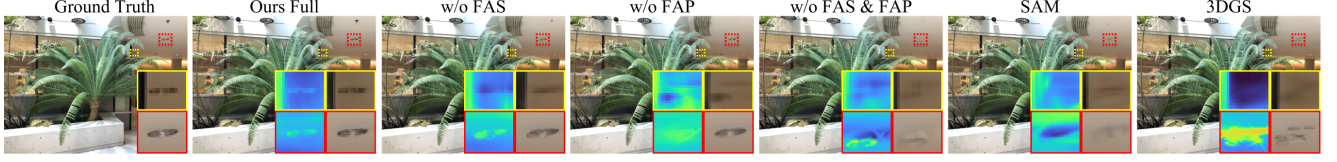


Figure 5. **Ablation study.** FAP denotes frequency-adaptive perturbation magnitude, FAS denotes frequency-adaptive sharpness weighting. “3DGS”, “SAM”, and “w/o FAS & FAP” produce inaccurate geometry (red box). All except “Ours Full” show blurry results (yellow box).

Table 2. **Ablation study.** We report averaged results over multiple runs. Standard deviations are omitted due to space constraints. FAP denotes frequency adaptive perturbation magnitude, FAS denotes frequency adaptive sharpness weighting, and GAP denotes per-Gaussian attribute perturbation.

3DGS + Components				LLFF (3 Views)		
SAM	GAP	FAP	FAS	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
✓	✓	✓	✓	20.783	.7197	.1965
✓	✓	✓	✗	20.560	.7116	.2023
✓	✓	✗	✓	20.570	.6980	.2142
✓	✓	✗	✗	20.390	.6977	.2174
✓	✗	✗	✗	20.198	.6958	.2095
✗	✗	✗	✗	19.810	.6790	.2145

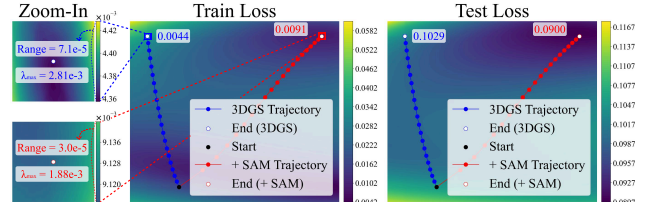
SAM [12], and our proposed method. For visualization, we project the high-dimensional parameter trajectories onto a 2D plane using Principal Component Analysis with parameters from both trajectories.

SAM converges to flatter minima than 3DGS. In Fig. 6.a, the loss range between the local maximum and minimum is $2.37\times$, and the measured sharpness is $1.49\times$ smaller. SAM also achieves a test loss of 0.0129 lower than 3DGS, narrowing the generalization gap from 0.0985 to 0.0809. However, our method converges to less flat minima than SAM. As shown in Fig. 6.b, the local loss range and measured sharpness of ours are $1.11\times$ smaller and $1.01\times$ smaller than 3DGS, respectively. Interestingly, our method achieves test loss of 0.0141 lower than 3DGS and further narrows the generalization gap to 0.0805, outperforming SAM in generalization.

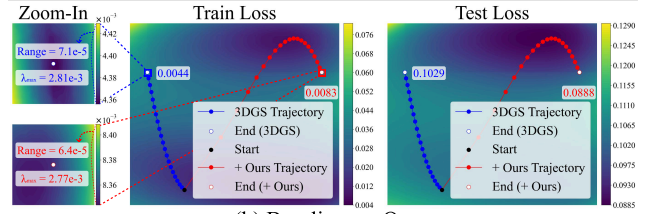
These results suggest that sharpness is not strictly correlated with generalization, supporting our hypothesis that sharpness of high-frequency pixel should be preserved. We further show that SAM tends to over-penalize high-frequency details, which leads to blurry results (Fig. 5).

4.3.3. Performance improvement by covisibility level

To examine where our method contributes to reconstruction quality, we measure AVGE across different covisibility levels. Following CoMapGS [20], we compute covisibility maps using MAST3R [31]. As shown in Tab. 3, the improve-



(a) Baseline vs. SAM



(b) Baseline vs. Ours

Figure 6. **Loss landscape visualization.** We compare the convergence of 3DGS and ours. Since the visualization yields a smoothed loss landscape, we also show a zoomed-in view near the convergence points of ours and the baseline. We measure sharpness as the maximum eigenvalue $\lambda_{\max}$ of the Hessian matrix [40, 60].

Table 3. **Performance by covisibility level on LLFF dataset.** Our method shows greater improvement with higher view sparsity.

Covisibility level	AVGE $\downarrow$		
	3DGS	+ Ours	Δ
Covisibility 3	.0483 $\pm$.0093	.0417 $\pm$.0080	- .0066 $\pm$.0043
Covisibility 2	.0757 $\pm$.0156	.0644 $\pm$.0137	- .0114 $\pm$.0067
Covisibility 1	.0948 $\pm$.0256	.0806 $\pm$.0232	- .0142 $\pm$.0074

ment of our method over the baseline 3DGS increases progressively as the covisibility decreases. This behavior aligns with our hypothesis that our method enhances generalization, particularly in under-constrained regions.

4.4. Application

4.4.1. Reducing computation cost of FASR

The main limitation of SAM and its follow-up work is that they compute the loss gradient twice at each step, theoretically doubling the training time. Consequently, recent work [21, 38, 61] has focused on improving efficiency. Among them, Zhou et al. [71] demonstrated that SAM operates efficiently during the final few training epochs.

Table 4. **Applying FASR at late training phase.** For efficient training, we apply our method during the later iterations, denoted as Ours-L. Results are averaged over multiple runs using an RTX A5000. Standard deviations are omitted for brevity.

Method	LLFF (3 Views)			
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	Time $\downarrow$
3DGS	19.810	.6790	.2145	85 sec
3DGS + Ours	20.783	.7197	.1965	237 sec
3DGS + Ours-L	20.591	.7102	.2054	98 sec

Table 5. **Applying FASR to online dynamic 3D Gaussians.** Our method is effective in dynamic scenarios under temporal sparsity, notably improving temporal consistency by reducing mTV.

Method	Neural 3D Video		
	PSNR $\uparrow$	SSIM $\uparrow$	mTV $\downarrow$
Yun et al. [SIGGRAPH'25]	32.542	.9486	.1109
+ Ours	32.622	.9497	.0989

Based on this finding, we apply our method during the last 12.5% of the total iterations, denoted as Ours-L. As shown in Tab. 4, Ours-L shows decreases of 0.192, 0.0095, and 0.0089 in PSNR, SSIM, and LPIPS, respectively compared to Ours, but still achieves improvements of 0.789, 0.0319, and 0.0097 over the baseline. Moreover, compared to the baseline, Ours increases the training time by $2.79\times$, Ours-L increases it by only $1.15\times$, making it more efficient.

The performance degradation of Ours-L occurs because 3DGS adaptively adjusts its number of learnable parameters through densification—a key difference from neural networks. This makes early training iterations influential, leading different results from Zhou et al. [71].

4.4.2. Improving generalization in temporal sparsity

Additionally, we extend our method to dynamic scenes, the Neural 3D Video dataset [34]. Specifically, we conduct experiments with an online configuration [14, 17, 33, 63] where observation is temporally sparse, meaning that we can only access the current frame in the sequentially processed video frames. We applied our method to Yun et al. [66] with 3DGStream [54] backbone. Following their protocol, we select the first frame 3D Gaussians with the highest PSNR for initialization and compute the masked total variation (mTV) to measure temporal consistency.

Our method improves temporal consistency and visual quality compared to the baseline (Tab. 5). This shows that our approach enhances generalization not only in the spatial domain but also in the temporal domain. Furthermore, Yun et al. [66] claims that one cause of temporal inconsistency is the inevitable noise in training datasets. Since SAM shows robustness on noisy training data, our finding aligns with this explanation.



Figure 7. **Comparison with FreeNeRF.** Our method corrects geometric inaccuracy of FreeNeRF.

Table 6. **Applying FASR to FreeNeRF.** Our intuition improves NeRF baseline.

Method	LLFF (3 views)		
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
FreeNeRF [CVPR'23]	19.523	.6063	.3103
+ Ours	19.584	.6182	.2983

4.4.3. Applying our intuition to NeRF

Despite our method is not directly designed for implicit representations such as NeRF [43] because a single parameter affects all pixels, we can alternatively apply our intuition to baselines that learn in a coarse-to-fine procedure [37, 64]. Specifically, we set a strong perturbation magnitude and regularization weight when the NeRF parameters learn low-frequency components, and gradually decrease these values as the model learns high-frequency details.

We applied this approach to FreeNeRF [64] and demonstrate that it improves both visual quality in novel-view synthesis (Fig. 7) and quantitative results (Tab. 6). Although the improvement over the NeRF baseline is smaller than that in the 3DGS baselines due to its alternative application, these additional gains nonetheless support our intuition and demonstrate its versatility across different 3D representations.

5. Conclusion

In this work, we present the first fundamental investigation linking loss landscape and generalization, thereby improving novel-view synthesis of 3D Gaussian Splatting [23], especially in sparse view reconstruction. We propose Frequency-Adaptive Sharpness Regularization (FASR), an optimization algorithm that reformulates Sharpness-Aware Minimization (SAM) [12] in a frequency-adaptive manner. This reformulation overcomes the limitation of SAM in reconstruction tasks, achieving generalization with fine detail preservation. Experiments demonstrate that FASR improves generalization and is easily applicable across diverse baselines, further extending to NeRF-based models and dynamic scenes in temporally sparse scenarios. We hope this work inspires the research community to explore the link between sharpness and generalization in reconstruction fields.

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Frequency-Adaptive Sharpness Regularization for Improving 3D Gaussian Splatting Generalization

Supplementary Material

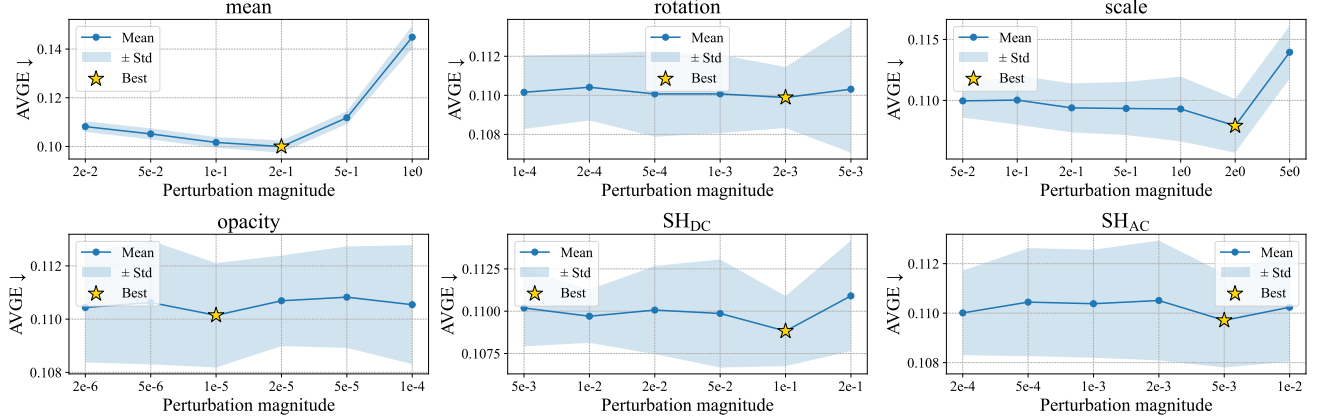


Figure 8. **Hyperparameter grid search, where each parameter is perturbed individually on the LLFF dataset.** Plots are means and standard deviations over 10 runs. We mark a star at the best hyperparameter value.

A. Hyperparameter search

In this section, we discuss selecting the hyperparameter neighborhood of radius ρ , i.e., the perturbation magnitude. Although our method scales ρ in a frequency-adaptive manner, ρ itself remains a core hyperparameter. We first search for the optimal perturbation magnitude by perturbing each parameter individually, as shown in Fig. 8.

However, the parameters are not independent and mutually influence each other. Consequently, the optimal ρ when perturbing all parameters simultaneously is typically smaller than the values found in this search. Therefore, we heuristically select smaller ρ values derived from this search as candidates and then fine-tune using this selection, perturbing all parameters simultaneously.

Selecting the optimal ρ remains a challenging problem in SAM-based methods. Recent studies, such as Wang et al. [57], explore learning the perturbation magnitude. Incorporating such approaches could be beneficial.

B. Perturbation per-attributes

To analyze the contribution of each Gaussian attribute to the overall performance, we conducted an ablation study. We apply our method only to a single attribute or by excluding a single attribute. As shown in Tab. 7, we observe that while all attributes contribute positively to the performance, the Gaussian mean is the dominant contributor to the performance gain.

This finding suggests that instead of perturbing all at-

Table 7. **Ablation study on the contribution of each Gaussian attributes.** The Gaussian mean is the dominant contributor to the performance improvement.

Method	LLFF (3 views)		
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
3DGS + Ours	20.783 $\pm$.300	.7197 $\pm$.0032	.1965 $\pm$.0034
w/o mean	20.053 $\pm$.228	.6903 $\pm$.0051	.2102 $\pm$.0040
w/o rotation	20.656 $\pm$.241	.7175 $\pm$.0046	.1976 $\pm$.0040
w/o scale	20.641 $\pm$.235	.7171 $\pm$.0049	.1987 $\pm$.0035
w/o opacity	20.628 $\pm$.218	.7175 $\pm$.0044	.1977 $\pm$.0033
w/o SH _{DC}	20.594 $\pm$.242	.7152 $\pm$.0052	.1972 $\pm$.0036
w/o SH _{AC}	20.599 $\pm$.205	.7169 $\pm$.0041	.1973 $\pm$.0034
w/ mean	20.561 $\pm$.268	.7147 $\pm$.0046	.1975 $\pm$.0037
w/ rotation	19.854 $\pm$.168	.6828 $\pm$.0029	.2110 $\pm$.0024
w/ scale	20.036 $\pm$.222	.6871 $\pm$.0048	.2093 $\pm$.0037
w/ opacity	19.888 $\pm$.188	.6803 $\pm$.0045	.2124 $\pm$.0033
w/ SH _{DC}	19.990 $\pm$.202	.6867 $\pm$.0048	.2115 $\pm$.0034
w/ SH _{AC}	19.914 $\pm$.186	.6827 $\pm$.0042	.2114 $\pm$.0033
3DGS	19.810 $\pm$.339	.6790 $\pm$.0078	.2145 $\pm$.0065

tributes simultaneously—which complicates the tuning of ρ due to mutual parameter influence as shown in Appendix A—applying our method exclusively to the Gaussian mean could be a simplified approach, significantly reducing hyperparameter search complexity while likely retaining a substantial portion of the performance improvement.

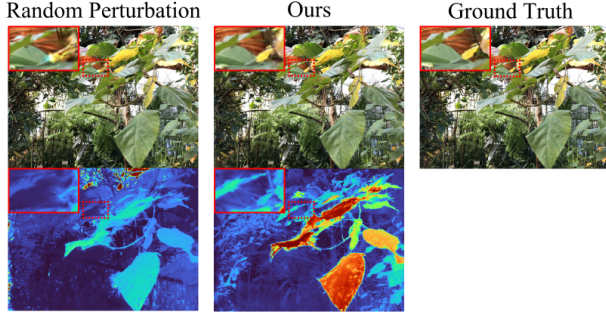


Figure 9. **Qualitative comparison of random perturbation and our method.** Random perturbation often introduce unexpected artifacts.

Method	LLFF (3 views)		
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
3DGS + Ours	20.783 $\pm$.300	.7197 $\pm$.0032	.1965 $\pm$.0034
w/o AP	20.263 $\pm$.289	.6990 $\pm$.0031	.2056 $\pm$.0027
w/ RP	20.174 $\pm$.207	.6987 $\pm$.0040	.2009 $\pm$.0032
3DGS	19.810 $\pm$.339	.6790 $\pm$.0078	.2145 $\pm$.0065

Table 8. **Quantitative comparison of random perturbation and adversarial perturbation.** AP denotes adversarial perturbation, and RP denotes random perturbation on Gaussian parameters.

C. Comparison to random perturbation

To validate our claims in Sec. 2.3 regarding the importance of adversarial perturbation, we compare our method with baselines that utilize random perturbation. For a fair comparison, we performed a grid search to find the optimal perturbation magnitude, same as Appendix A. As shown in Fig. 9, random perturbations often introduce unexpected artifacts.

Furthermore, we also compare a variant where the gradient is computed at a randomly perturbed position instead of the local maximum found adversarially. As shown in Tab. 8, all random perturbation methods improve 3DGS. However, their performance gains are significantly smaller than ours, indicating the importance of using adversarial perturbation.

D. Analysis of Application Phase

We provide a detailed study on the application timing of our method, expanding the setup of Sec. 4.4.1. In the main experiment, following Zhou et al. [71], we applied our method only during the final 12.5% of the total training iterations. Extending this, we conducted experiments applying our method during the last $N \cdot 10\%$ of the total iterations. As shown in Fig. 10, fully applying our method yields the best performance, and the performance degrades as the application phase is later in the training process.

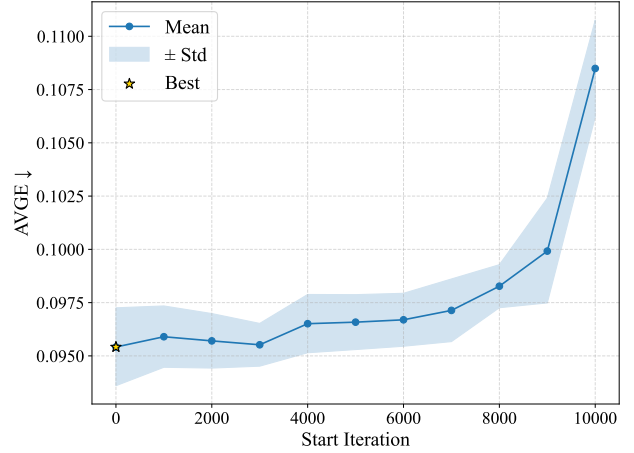


Figure 10. **Ablation study on the application phase.** We show the improvement of applying our method during the last $N \cdot 10\%$ of total training iterations.

E. LoG for frequency estimation.

We calculate the local frequency at each pixel of the input image employing a multi-scale analysis inspired by blob detection using the Laplacian of Gaussian (LoG) [36]. For the grayscale image, we calculate the absolute LoG response at each candidate scale. We capture the optimal scale at each pixel by finding the first significant increase in its response curve that exceeds the threshold. The optimal scale is the smallest one that precedes such a rise; otherwise, we assign the maximum candidate scale. This process results in an optimal scale map $\Gamma^v \in \mathbb{R}^{H \times W}$ at view v , with H and W denoting the height and width of the image, respectively. The local frequency is inversely proportional to this selected scale.

F. Implementation detail

For all methods except 3DGS, we use the official repository. Since the official 3DGS does not target sparse-view reconstruction, we refactor it by referring to CoRGS and DropGaussian. NexusGS does not provide optical flow for MipNeRF-360; we compute the flow using FlowFormer++ [51]. When applying regularization, we back-propagate the regularization term after computing the gradient of FASR. For each dataset, we used the same values of ρ_θ and γ for all baselines except NexusGS, where we set them to $0.1 \times$ the values used for other baselines due to its use of denser per-pixel Gaussians. When applying our method in the deformation process, we compute the FASR gradient on the deformed Gaussians and propagate it back to the deformation network. We update the residual map of Yun et al. [66] during the ascent step and the Gaussians during the descent step.

Table 9. **Per-scene quantitative results on LLFF and MipNeRF-360 dataset.** We report AVGE (lower is better) of each scene.

Method	LLFF (3 views)							
	fern	flower	fortress	horns	leaves	orchids	room	trex
3DGS	.0950 $\pm$.01790	.1147 $\pm$.00180	.0885 $\pm$.00390	.1276 $\pm$.00240	.1362 $\pm$.00170	.1769 $\pm$.00190	.0753 $\pm$.00160	.0749 $\pm$.00210
+ Ours	.0765 $\pm$.00080	.1066 $\pm$.00140	.0781 $\pm$.00530	.1044 $\pm$.00280	.1170 $\pm$.00130	.1605 $\pm$.00190	.0686 $\pm$.00190	.0712 $\pm$.00710
CoR-GS	.0806 $\pm$.00080	.1088 $\pm$.00290	.0754 $\pm$.00160	.1205 $\pm$.00120	.1427 $\pm$.00310	.1735 $\pm$.00200	.0723 $\pm$.00130	.0699 $\pm$.00440
+ Ours	.0724 $\pm$.00090	.1014 $\pm$.00220	.0683 $\pm$.00190	.1075 $\pm$.00200	.1326 $\pm$.00240	.1603 $\pm$.00100	.0695 $\pm$.00100	.0614 $\pm$.00340
DropGaussian	.0791 $\pm$.00120	.1056 $\pm$.00210	.0796 $\pm$.00420	.1186 $\pm$.00270	.1284 $\pm$.00190	.1650 $\pm$.00160	.0715 $\pm$.00150	.0687 $\pm$.00190
+ Ours	.0717 $\pm$.00100	.1048 $\pm$.00170	.0784 $\pm$.00340	.1101 $\pm$.00340	.1176 $\pm$.00140	.1536 $\pm$.00170	.0701 $\pm$.00200	.0657 $\pm$.00260
NexusGS	.0893 $\pm$.00030	.0981 $\pm$.00020	.0498 $\pm$.00030	.0909 $\pm$.00050	.1072 $\pm$.00030	.1383 $\pm$.00060	.0755 $\pm$.00060	.0810 $\pm$.00050
+ Ours	.0848 $\pm$.00050	.0989 $\pm$.00060	.0475 $\pm$.00090	.0897 $\pm$.00070	.0993 $\pm$.00060	.1354 $\pm$.00040	.0696 $\pm$.00030	.0741 $\pm$.00060
SE-GS	.0720 $\pm$.00040	.1052 $\pm$.00180	.0662 $\pm$.00130	.1104 $\pm$.00180	.1392 $\pm$.00230	.1707 $\pm$.01440	.0635 $\pm$.00220	.0596 $\pm$.00180
+ Ours	.0675 $\pm$.00060	.0992 $\pm$.00160	.0661 $\pm$.00820	.1028 $\pm$.00160	.1310 $\pm$.00160	.1546 $\pm$.00170	.0629 $\pm$.00120	.0564 $\pm$.00080
Method	MipNeRF-360 (12 views)							
	bicycle	bonsai	counter	garden	kitchen	room	stump	
3DGS	.1763 $\pm$.00400	.1408 $\pm$.00400	.1464 $\pm$.00250	.1302 $\pm$.00190	.1052 $\pm$.00160	.1078 $\pm$.00110	.2347 $\pm$.00280	
+ Ours	.1714 $\pm$.00120	.1285 $\pm$.00240	.1350 $\pm$.00110	.1240 $\pm$.00130	.1017 $\pm$.00160	.1021 $\pm$.00460	.2253 $\pm$.00640	
CoR-GS	.1734 $\pm$.00340	.1262 $\pm$.00250	.1310 $\pm$.00120	.1297 $\pm$.00370	.1054 $\pm$.00360	.0931 $\pm$.00230	.2214 $\pm$.00770	
+ Ours	.1733 $\pm$.00170	.1131 $\pm$.00230	.1229 $\pm$.00060	.1287 $\pm$.00290	.1044 $\pm$.00320	.0993 $\pm$.00440	.2073 $\pm$.00950	
DropGaussian	.1646 $\pm$.00110	.1301 $\pm$.00210	.1356 $\pm$.00140	.1243 $\pm$.00110	.0979 $\pm$.00270	.1026 $\pm$.00500	.2187 $\pm$.00410	
+ Ours	.1613 $\pm$.00310	.1286 $\pm$.00070	.1309 $\pm$.00100	.1218 $\pm$.00200	.1027 $\pm$.00210	.1003 $\pm$.00440	.2171 $\pm$.01180	
NexusGS	.1818 $\pm$.00100	.1308 $\pm$.00190	.1563 $\pm$.00130	.1249 $\pm$.00030	.0941 $\pm$.00070	.1506 $\pm$.00380	.2352 $\pm$.00240	
+ Ours	.1739 $\pm$.00100	.1278 $\pm$.00060	.1531 $\pm$.00160	.1243 $\pm$.00040	.0921 $\pm$.00090	.1421 $\pm$.00450	.2299 $\pm$.00210	
SE-GS	.1624 $\pm$.00350	.1179 $\pm$.00300	.1251 $\pm$.00370	.1207 $\pm$.00160	.1107 $\pm$.00370	.0875 $\pm$.00380	.2014 $\pm$.00830	
+ Ours	.1599 $\pm$.00690	.1105 $\pm$.00300	.1217 $\pm$.00170	.1179 $\pm$.00250	.1092 $\pm$.00350	.0973 $\pm$.00090	.1941 $\pm$.00480	

G. More results

We report the quantitative results of each scene in Tab. 9.
Our method method outperforms the baseline in most cases.